

Consultas PostGis

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Geomática II



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Pantallazos sentencias SQL

1. Localidad con mayor número de caracteres

The screenshot shows a SQL query editor with the following code:

```
1  -- Localidad con nombre mas largo
2  SELECT "LocNombre"
3  FROM "Loca"
4  ORDER BY LENGTH("LocNombre") DESC
5  LIMIT 1;
```

Below the query editor, the 'Data Output' tab is active, showing a table with one row of results:

	LocNombre character varying (50)
1	RAFAEL URIBE URIBE

2. Localidad con mayor área y el área Ha. Area calculada

The screenshot shows a SQL query editor with the following code:

```
1  SELECT
2      "locnombre",
3      ST_Area("geom") / 10000 AS area_ha
4  FROM "Loca"
5  ORDER BY area_ha DESC
6  LIMIT 1;
```

Below the query editor, the 'Data Output' tab is active, showing a table with one row of results:

	locnombre character varying (50)	area_ha double precision
1	SUMAPAZ	78034.94333402446

3. Localidad con menor área y el área Ha. Area calculada

The screenshot shows a SQL query editor with a dark theme. The 'Query' tab is active, displaying a SQL query. Below the query, the 'Data Output' tab shows the results of the query. The query is as follows:

```
1 SELECT
2     "locnombre",
3     ST_Area("geom") / 10000 AS area_ha
4 FROM "loca"
5 ORDER BY area_ha ASC
6 LIMIT 1;
```

The 'Data Output' tab shows a table with two columns: 'locnombre' (character varying (50)) and 'area_ha' (double precision). The first row of data is:

	locnombre	area_ha
1	CANDELARIA	205.85930624811797

4. Segmento de red férrea más largo y su distancia en metros

The screenshot shows a SQL query editor with a dark theme. The 'Query' tab is active, displaying a SQL query. Below the query, the 'Data Output' tab shows the results of the query. The query is as follows:

```
6 SELECT
7     "nombre" AS segmento,
8     ST_Length("geom") AS longitud_m
9 FROM "red_ferrea"
10 ORDER BY longitud_m DESC
11 LIMIT 1;
```

The 'Data Output' tab shows a table with two columns: 'segmento' (character varying (250)) and 'longitud_m' (double precision). The first row of data is:

	segmento	longitud_m
1	Regiotram del occidente	40457.23188946946

5. SRID de la capa localidad

Query Query History

```
8
9 SELECT ST_SRID("geom") AS srid
10 FROM "loca"
11 LIMIT 1;
12
13
```

Data Output Messages Notifications

	srid integer
1	3116

6. Elemento (relacionar capa, ID, distancia) más largo entre red metro y red férrea. Distancia calculada

```
1 WITH elementos_combinados AS (
2     SELECT
3         'Red Metro' AS capa,
4         id,
5         ST_Length(geom) AS distancia_metros
6     FROM "Red_metro"
7
8     UNION ALL
9
10    SELECT
11        'Red Ferrea' AS capa,
12        id,
13        ST_Length(geom) AS distancia_metros
14    FROM "Red_ferrea"
15 )
16 SELECT
17     capa,
18     id,
19     distancia_metros
20 FROM elementos_combinados
21 ORDER BY distancia_metros DESC
22 LIMIT 1;
```

Data Output Messages Notifications

	capa text	id integer	distancia_metros double precision
1	Red Ferrea	1	40457.23188946946

7. Localidad(es) que no interseca(n) ningún elemento de red_ferrea

```

1  SELECT
2      l.id,
3      l."LocNombre" AS nombre_localidad,
4      l."LocCodigo" AS codigo
5  FROM "Loca" l
6  WHERE NOT EXISTS (
7      SELECT 1
8      FROM "Red_ferrea" rf
9      WHERE ST_Intersects(l.geom, rf.geom)
10 );

```

	id [PK] integer	nombre_localidad character varying (50)	codigo character varying (2)
1	1	ANTONIO NARIÑO	15
2	2	TUNJUELITO	06
3	3	RAFAEL URIBE URIBE	18
4	4	CANDELARIA	17
5	9	SUMAPAZ	20
6	13	SAN CRISTOBAL	04
7	14	USME	05
8	19	ENGATIVA	10
9	20	SUBA	11

8. Trazado troncal de transmilenio que cruza por 5 o más localidades y cuales son los trazados y las localidades que cruza.

```

1  SELECT
2      tt.id AS id_troncal,
3      tt."NOMBRE_TRA" AS nombre_troncal,
4      COUNT(DISTINCT l.id) AS num_localidades,
5      STRING_AGG(DISTINCT l."LocNombre", ', ' ORDER BY l."LocNombre") AS localidades
6  FROM "Trazado_troncal_transmilenio" tt
7  JOIN "Loca" l ON ST_Intersects(ST_SetSRID(tt.geom, 3116), ST_Transform(l.geom, 3116))
8  GROUP BY tt.id, tt."NOMBRE_TRA"
9  HAVING COUNT(DISTINCT l.id) >= 5
10 ORDER BY num_localidades DESC;

```

	id_troncal integer	nombre_troncal character varying (100)	num_localidades bigint	localidades text
1	4	NQS	5	BARRIOS UNIDOS, CHAPINERO, LOS MARTIRES, PUENTE ARANDA, TEUSAQUILLO
2	7	CARRERA 10	5	ANTONIO NARIÑO, CANDELARIA, RAFAEL URIBE URIBE, SAN CRISTOBAL, SANT...
3	9	CARACAS CENTRO	5	BARRIOS UNIDOS, CHAPINERO, LOS MARTIRES, SANTA FE, TEUSAQUILLO
4	10	CALLE 26	5	ENGATIVA, FONTIBON, LOS MARTIRES, SANTA FE, TEUSAQUILLO
5	11	AUTOPISTA SUR	5	BOSA, CIUDAD BOLIVAR, KENNEDY, PUENTE ARANDA, TUNJUELITO
6	12	CARACAS SUR	5	ANTONIO NARIÑO, LOS MARTIRES, RAFAEL URIBE URIBE, SANTA FE, TUNJUELIT...

9. Descargar de -Bogotá Datos abiertos- la capa “Comando de Atención Inmediata. Bogota D.C.” Localidad con mayor y menor número de CAI, Indicar nombre de la localidad y cantidad.

```
1 WITH cai_por_localidad AS (  
2     SELECT  
3         l."LocNombre" as localidad,  
4         COUNT(c.id) as cantidad_cai  
5     FROM "Loca" l  
6     LEFT JOIN "CAI" c ON ST_Intersects(l.geom, c.geom)  
7     GROUP BY l."LocNombre"  
8 ),  
9 max_min AS (  
10     SELECT  
11         MAX(cantidad_cai) as max_cai,  
12         MIN(cantidad_cai) as min_cai  
13     FROM cai_por_localidad  
14 )  
15 -- Localidad con mayor número de CAI  
16 SELECT  
17     'Mayor' as tipo,  
18     cpl.localidad,  
19     cpl.cantidad_cai  
20 FROM cai_por_localidad cpl  
21 WHERE cpl.cantidad_cai = (SELECT max_cai FROM max_min)  
22  
23 UNION ALL  
24  
25 -- Localidad con menor número de CAI  
26 SELECT |  
27     'Menor' as tipo,  
28     cpl.localidad,  
29     cpl.cantidad_cai  
30 FROM cai_por_localidad cpl  
31 WHERE cpl.cantidad_cai = (SELECT min_cai FROM max_min);
```

Data Output Messages Notifications



	tipo	localidad	cantidad_cai
	text	character varying (50)	bigint
1	Mayor	KENNEDY	15
2	Men...	SUMAPAZ	0

10. Cuál es el % de área de cobertura de atención en Bogotá de los CAI, con una zona de influencia de 200 m2. El área de Bogotá en km2 es de 1776. En los buffer ignorar el dissolve por solapamiento.

```
1 WITH
2 -- Crear buffers de 200m para cada CAI
3 buffers_cai AS (
4     SELECT ST_Buffer(geom, 200) as geom_buffer
5     FROM "CAI"
6 ),
7 -- Unir todos los buffers (sin disolver solapamientos)
8 area_cobertura AS (
9     SELECT ST_Union(geom_buffer) as geom_cobertura
10    FROM buffers_cai
11 ),
12 -- Calcular área de cobertura en km²
13 area_calculada AS (
14     SELECT
15         ST_Area(geom_cobertura) / 1000000 as area_cobertura_km2
16     FROM area_cobertura
17 )
18 -- Calcular porcentaje sobre el área total de Bogotá (1776 km²)
19 SELECT
20     area_cobertura_km2,
21     1776 as area_bogota_km2,
22     (area_cobertura_km2 / 1776) * 100 as porcentaje_cobertura
23 FROM area_calculada;
```

Data Output

Messages

Notifications

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SQL

	area_cobertura_km2 double precision 🔒	area_bogota_km2 integer 🔒	porcentaje_cobertura double precision 🔒
1	19.103244331818793	1776	1.0756331267915986